Table S4. *TSHR* and *IGF1R* tissue expression extension

A. Availability

Gene / tissue	GTEx N	Instrument (F)	Status
TSHR / Thyroid	574	—	No instrument (minimum eQTL $P = 9.60\times10^{-8}$)
IGF1R / Thyroid	574	rs1815009 (F = 61.1)	Estimable
TSHR / Cultured fibroblasts	483	—	Not tested by GTEx in this tissue
IGF1R / Cultured fibroblasts	483	rs6598538 (F = 33.8)	Estimable
TSHR / Subcutaneous adipose	581	rs12101255 (F = 38.5)	Estimable
IGF1R / Subcutaneous adipose	581	—	No instrument (minimum eQTL $P = 3.56\times10^{-7}$)
TSHR / Visceral adipose	469	rs74064796 (F = 32.1)	Estimable
IGF1R / Visceral adipose	469	—	No instrument (minimum eQTL $P = 7.76\times10^{-6}$)
TSHR / Whole blood	670	—	No instrument (minimum eQTL $P = 2.46\times10^{-4}$)
IGF1R / Whole blood	670	—	No instrument (minimum eQTL $P = 5.42\times10^{-5}$)

B. Estimates

Gene / tissue	Dataset	Family	OR (95% CI)	*P* value	Below family threshold	Direction vs blood
TSHR / Subcutaneous adipose	BBJ GD	Secondary	0.46 (0.38–0.55)	1.92×10^{-15}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GD	Primary	0.36 (0.31–0.43)	1.28×10^{-34}	Yes	Same
TSHR / Subcutaneous adipose	UKB hyperthyroidism	Secondary	0.31 (0.27–0.37)	2.86×10^{-42}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO	Secondary	0.45 (0.32–0.63)	5.76×10^{-6}	Yes	Same
TSHR / Subcutaneous adipose	FinnGen GO strict	Secondary	0.51 (0.35–0.74)	3.67×10^{-4}	Yes	Same
TSHR / Visceral adipose	BBJ GD	Secondary	0.56 (0.46–0.67)	2.64×10^{-10}	Yes	Same
TSHR / Visceral adipose	FinnGen GD	Primary	0.50 (0.43–0.58)	4.09×10^{-19}	Yes	Same
TSHR / Visceral adipose	UKB hyperthyroidism	Secondary	0.39 (0.34–0.45)	9.19×10^{-38}	Yes	Same
TSHR / Visceral adipose	FinnGen GO	Secondary	0.69 (0.49–0.98)	0.0368	No	Same
TSHR / Visceral adipose	FinnGen GO strict	Secondary	0.78 (0.54–1.13)	0.186	No	Same
IGF1R / Thyroid	BBJ GD	Secondary	1.23 (0.90–1.68)	0.200	No	Same
IGF1R / Thyroid	FinnGen GD	Primary	0.76 (0.56–1.02)	0.0692	No	Opposite
IGF1R / Thyroid	UKB hyperthyroidism	Secondary	1.33 (0.99–1.80)	0.0594	No	Same
IGF1R / Thyroid	FinnGen GO	Secondary	1.02 (0.54–1.94)	0.945	No	Same
IGF1R / Thyroid	FinnGen GO strict	Secondary	1.23 (0.62–2.44)	0.549	No	Same
IGF1R / Cultured fibroblasts	BBJ GD	Secondary	0.91 (0.70–1.20)	0.511	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GD	Primary	1.11 (0.89–1.38)	0.359	No	Same
IGF1R / Cultured fibroblasts	UKB hyperthyroidism	Secondary	1.03 (0.82–1.29)	0.815	No	Same
IGF1R / Cultured fibroblasts	FinnGen GO	Secondary	0.78 (0.49–1.25)	0.301	No	Opposite
IGF1R / Cultured fibroblasts	FinnGen GO strict	Secondary	0.78 (0.47–1.30)	0.341	No	Opposite

Post hoc fixed-panel extension using GTEx v8; tissue colocalization was not performed. A, availability of the ten gene–tissue combinations; F statistics are for the single retained instrument. ‘Not tested’ means the gene is not among the genes GTEx tested in that tissue. GTEx whole blood (n = 670) yielded no instrument at the fixed threshold, whereas the main analysis used eQTLGen (up to 31,684 participants). B, all 20 estimates. The primary family is the ten FinnGen GD comparisons ($P < 0.005$); the secondary family is the 40 comparisons with the other datasets ($P < 0.00125$); unestimable combinations stay in both denominators. ORs are per unit of inverse-normal-transformed tissue expression and are not comparable in size with the blood ORs. With one instrument, each *P* value equals that variant’s disease-association z-test *P* value (UKB source BOLT-LMM *P* values differ slightly). The adipose *TSHR* instruments (rs12101255, rs74064796; 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and re-express the regional association rather than providing independent evidence. Cultured fibroblasts and adipose tissue are not orbital tissue. Direction, compared with the blood estimate in the same dataset. CI, confidence interval; eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; OR, odds ratio; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.